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PI Name:	Mitchell, Cary Ph.D.		
Project Title:	(NSCORT) for Advanced Life Support at Purdue/Howard/AAMU - Solid Waste Resource Recovery		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
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Human Research Program Elements:	None		
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Zip Code:	47907-2010	**Congressional District:**	4
Comments:			
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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Alleman , Jim (Purdue Univesity) Glass, Charles (Howard University) Turco, Ron (Purdue University) Nies, Larry (Purdue University) Mitchell, Cary (Purdue University)		
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Task Description:

The Solid Waste Resource Recovery Focus Area consists of three projects and the task descriptions are as follows:
SOLID-PHASE THERMOPHILIC AEROBIC REACTOR (STAR) PROCESSING OF FECAL, FOOD, AND PLANT RESIDUES The Solids Thermophilic Aerobic Reactor (STAR) was investigated for the initial treatment of all biodegradable solid wastes that would be generated in a crewed mission, including inedible plant wastes, paper, fecal matter, and food wastes. STAR operates similarly to the autothermal thermophilic aerobic digester (ATAD) currently in use in a number of wastewater treatment plants. The projected advantages of the STAR system include a reduced retention time, increased rapid pathogen inactivation, lower reactor volume requirements, and ease of automation as compared to other biological waste treatment systems.

THE PABLO PROJECT: PLANT-BASED ANAEROBIC-AEROBIC BIOREACTOR LINKED OPERATION

This project focuses on the adaptation and use of anaerobic-aerobic technologies for the treatment of three types of biological waste (plant residues, greywater and human waste) for the Mars colony and for use in the production of Earth-based biofuels. The primary goal is the anaerobic conversion of a variety of plant waste to CH4 (Methane). The aerobic portion of the project is for conversion recalcitrant materials to form materials that can be further converted to CH4. This is a novel approach that will have applications in Earth-based as well as space. For the Mars portion of the project we are interested the use of dried plant residues and other organic materials grown within the space colony. Consideration is also given to the impact of long-term storage of materials. For the Earth-based biofuels generation, we are interested in wet waste materials (food processing) as well as dried residues. While we are presently perusing methane production from plant waste, the long-term goals is to develop a dual systems, one to handle the combination of plant waste and greywater and one to handle the human waste. While both will result in the formation of CH4, the plant waste-grey water system Plant-based Anaerobic-aerobic Bioreactor Linked Operation (PABLO) will also be used to develop an organic plant growth media similar to compost. The human waste treatment system Waste-ABLO (WABLO) uses the same approach but the resulting biosolids are not used in food production. In both cases, anaerobically processed waste streams will provide an energy source based on a renewable source and will result in a reduction in the mass of plant waste present. In the long-term the anaerobic digestor will be a source of methane (leading to small scale electricity generation), produce water, treat waste materials while conserving O2 and act as a heat sink as optimal digestor operation is achieved with an elevated temperature. Advanced biological treatment may be able to convert CH4 to methanol that can be used directly in fuel cells. The PABLO solid waste materials will be an ideal bedding material for the production of mushrooms and any other plant types that are difficult to grow in liquid systems. In order to maintain a high quality food, at no time do we propose using biosolids from WABLO in food production. These anaerobic digestor systems will provide a robust treatment technology that will allow a rapid and consistent conversation of waste to energy and in the case of PABLO needed secondary plant growth materials.

NITROGEN CYCLING IN ADVANCED LIFE SUPPORT SYSTEMS

Currently, NASA employs physiochemical technology to purify water for “short” space journeys. Conversely, this process is expensive and less than ideal for long voyages where optimally a significant portion of the waste products must be regenerated, especially the recovery of potable water from wastewater. If NASA plans to sustain long-term human voyages into space, to achieve this goal, most of the waste generated should be recycled. This project will contribute to the knowledge base focusing on sustaining human life for longer periods of time in space, specifically a Mars Mission. This research focused on converting a concentrated ammonia-rich liquid, which includes trace organics and hydrogen sulfide from the Sequential Thermophilic Aerobic Reactor (STAR) Gas Scrubber, into a form that would be conducive for downstream processes of gray water treatment and plant growth. We identified the optimum zeolite to effectively treat ammonia rich wastewater and convert it to a form appropriate for plant uptake, while developing a regenerative process for the zeolite.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Further development of biological methods for converting organic wastes into fuels, chemicals, and other useful products.
Add to knowledge about low-flush toilet operations

Gain a greater understanding for recycling ammonia-rich wastewater streams.

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Research Progress

1. Oxygen Transfer, Solids Loading, and Sensitivity Analysis: The reactor design was further improved and evaluated over the last year. An evaluation of oxygen transfer, solids loadings, and a sensitivity analysis were completed on the final design.

Oxygen transfer in the system was considerably improved by the enhancements to the reactor design. An 8% solids loading with a 9 day HRT (Hydro Residence Time) was selected as the optimum operating scenario for further research in the system as currently configured. This selection was deemed appropriate for further research goals; however, mission architecture and post-treatment technology selection would dictate the outcome of this decision for long-term missions. A sensitivity analysis was completed on the reactor to determine the most statistically significant parameters for degradation of solid waste and the concomitant generation of CO2. Significant parameters were determined to be pH, ORP, temperature, and time.

2. Nitrogen Pathways and Attainment of Stable Cycling: In summation, the nitrogen pathway is expected to proceed as follows: Initially, the influent sources of nitrogen are dominated by particulate organic nitrogen as part of cellular

composition, both as microbial biomass and feedstock particulate matter. This particulate organic nitrogen is rapidly released into the bulk solution as the cellular solids lyse and decay and proteins are denatured. The soluble organic nitrogen, largely from amino acid degradation, is hydrolyzed into soluble inorganic nitrogen. Finally, the inorganic nitrogen strives for equilibrium in the bulk solution via stripping (primarily NH3, potentially as NO2), and as soluble inorganic nitrogen. Increases in pH and alkalinity are largely due to stripping of CO2 and NH3, and the formation of carbonates.

In the final design of this investigation, the pilot-scale reactor was operated using water to dilute the influent raw waste stream to the desired solids level. Filtrate recycling would contribute to operational cost reduction by reducing or eliminating the need for addition of clean water to the system to increase moisture levels, and was also proven to increase degradation in the system. Another option for investigation would be the utilization of grey water from the system as a wetting agent.

The PABLO project: Plant-based Anaerobic-aerobic Bioreactor Linked Operation

The project is divided into three tasks:

Task 1. Use an anaerobic-aerobic bioreactor system to optimize production of methane from plant-based waste materials and for the recovery of high-quality biosolids.

Task 2. Methods for the optimization of the PABLO biosolids for use in fungal and plant cultivation will be investigated.

Task 3. The objective is use a bioreactor system to optimize production of methane from human waste materials and to provide a means of drying and storing the material.

Progress to date:

Work has been concentrated on establishing testing protocols for different waster materials. We are using a simple digester system to evaluate gas production. This has allowed us to begin to evaluate plant materials in single entries or combination applications. Our data suggests that a variety of plant materials are capable of being used to produce CH4. (Methane) Of note is the finding that some materials produce a lot less gas than others and this should factor into the discussion making process that goes into meal selection for space flight. For example, carrots are surpassed by vines from sweet potato and a combination of tomato and spinach materials. We are now concentrated on addressing gas production from plant residual materials as these will be in abundance at the space-base but we are also considering plant/fruit waste as this will be available on earth.

Task Progress:

Ongoing work is assessing combinations of fruit and vine waste residues so that we can provide input towards meal planning and cropping sequences. That is, given a choice of two foods stuffs to grow, our input will provide information as to the one that will provide the best level of energy production as a waste material. We are also looking at reactor design. The contribution has important impacts to the space program as well as to Earth applications. For example, which combinations of materials will provide the most significant level of gas production? This has been largely overlooked as a contributor to the success of bioenergy programs. We are also looking at biological approaches to converting CH4 to methanol. The advantage is that methanol can be easily stored and can be used directly in fuel cells. While this will have applications in space, our major program is aimed at earth applications.

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PROJECT GOALS AND OBJECTIVES

• To effectively treat ammonia rich wastewaters from various process streams inside of the ALS system.

• To convert concentrated ammonia to a form for uptake by plants or other unit operations in ALS.

RESEARCH PROGRESS

Adsorption of Ammonia on to Zeolites. Natural zeolites have been researched extensively for their use in nitrogen removal from wastewater from industrial and domestic waste sources. At concentrations greater that 100 mg/L NH3-N the ion exchange process, using the ammonium selective ion exchange media, become feasible for use in comparison to biological nitrification and other methods such as ammonia stripping and breakpoint chlorination followed by treatment on activated carbon.

Ammonium removal by clinoptilolite has been found to be a cost effective method in wastewater treatment over the last three decades. Adsorption of ammonium is well reported for the two zeolite types, clinoptilolite and chabazite, and was carried out in both batch and column reactors to evaluate the effective adsorption capacity of these zeolites at high concentrations of ammonia in simulated wastewater.

The chabazite was found to be the most stable and had a maximum adsorption capacity at equilibrium (32.73 mg NH3-N/ gzeolite) as compared to the commonly used clinoptilolite. A column study was also performed with synthetic wastewater to determine the breakthrough curves for the three zeolites. The chabazite was found to outperform the other forms of zeolite which was consistent with the results of the batch reactors. A pretreatment process involving an acid treatment may result in dissolving alkali impurities which can block pores and channels within the zeolitic framework, hence promoting adsorption deep within the pores. However, zeolites with lower Si/Al ratio have been found to be

	unstable at lower pH, probably due to the loss of aluminum from the framework and thus, resulting in lower adsorption capacity. Another pretreatment aimed at increasing the ammonia adsorption capacity of the zeolite was treatment with sodium hydroxide. Heating in the presence of salt for varying time periods did not improve the adsorption capacity. The retention capacity of one hour heat treated chabazite was the best among all the other pretreated samples. Nitrification and Denitrification. In the phase III Lunar-Mars Support Test project ammonium ions were removed using a column in series system packed with clinoptilolite, a naturally occurring zeolite material. Clinoptilolite removes ammonium ions through ion exchange with calcium. The system worked well, however the clinoptilolite was never regenerated and the nitrogen sorbed was never returned to the system. This would not cause a problem in a 91-day experiment, or any other short term mission, however over time nitrogen would accumulate in this phase and an unstable system would result. In this project a nitrification/denitrification reactor is being used to regenerate the clinoptilolite and return a liquid product that can be used as a nutrient supplement for the plant systems in the ALS.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Whitaker DR, Alleman JE. "Lignin Degradation in Solid Waste Treatment Systems." Habitation Conference, American Institute of Aeronautics and Astronautics, Orlando, FL, February 2006. Habitation 2006 Feb;10(3-4):189. , Feb-2006
Articles in Peer-reviewed Journals	Glass CC, Ressa C. "Enhancement of Ammonium Adsorption Capacity with Pretreated Natural Zeolites." Habitation, Accepted with Major Revisions, 2006. , Jul-2006
Papers from Meeting Proceedings	Whitaker DR, Staton KL, Alleman JE, Lane JW. "Loading Balance and Influent pH in a Solids Thermophilic Aerobic Reactor." International Conference on Environmental Systems, Society of Automotive Engineers, Aerospace Division, Rome, Italy, July 2005. International Conference on Environmental Systems, Society of Automotive Engineers, Paper # 2005-01-2982, 2005. , Jul-2005
Papers from Meeting Proceedings	Kuo Y, Whitaker D, Chiu G, Alleman J. "System Level Design and Initial Equivalent System Mass Analysis of a Solid-Phase Thermophilic Aerobic Rector for Advanced Life Support Systems." International Conference on Environmental Systems, Society of Automotive Engineers, Aerospace Division, Rome, Italy, July 2005. International Conference on Environmental Systems, Society of Automotive Engineers, Paper # 2005-01-2983, 2005. , Jul-2005
Papers from Meeting Proceedings	Whitaker DR, Alleman JE. "Thermophilic Aerobic Solid Waste Processing for Long-term Crewed Missions." Earth and Space 2006, 10th ASCE Conference on Engineering, Construction, and Operations in Challenging Environments, Houston, TX, March 2006. Earth and Space 2006, 10th ASCE Conference on Engineering, Construction, and Operations in Challenging Environments, 2006. , Mar-2006